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BLSN Report ? P3 Orion /* ?? Design Tokens: Light (default)
???????????????????????????????? */ :root, [data-theme="light"] { --bg-page:
#f8fafc; --bg-card: #ffffff; --bg-muted: #f1f5f9; --bg-header: #f8fafc;
--border: #e2e8f0; --text-1: #0f172a; --text-2: #475569; --text-3: #94a3b8;
--accent: #6366f1; --accent-bg: #eef2ff; --pass-text: #15803d; --pass-bg:
#f0fdf4; --warn-text: #b45309; --warn-bg: #fffbeb; --fail-text: #b91c1c; --fail-
bg: #fef2f2; --shadow: 0 1px 3px 0 rgb(0 0 0 / 0.07); } /* ?? Design Tokens:
Dark ???????????????????????????????????????????? */ [data-theme="dark"] {
--bg-page: #0f172a; --bg-card: #1e293b; --bg-muted: #1e293b; --bg-header:
#1e293b; --border: #334155; --text-1: #f1f5f9; --text-2: #94a3b8; --text-3:
#64748b; --accent: #818cf8; --accent-bg: #1e1b4b; --pass-text: #4ade80; --pass-
bg: #052e16; --warn-text: #fbbf24; --warn-bg: #1c1400; --fail-text: #f87171;
--fail-bg: #1c0505; --shadow: 0 1px 3px 0 rgb(0 0 0 / 0.3); } /* ?? Density
Scale ???????????????????????????????????????????????????? */ /* Default
(comfortable) spacing: these vars are overridden per density. */ :root { --d-py:
0.75rem; /* table cell vertical padding */ --d-px: 1rem; /* table cell
horizontal padding */ --d-gap: 1rem; /* grid/flex gap */ --d-card: 1.5rem; /*
card padding */ } [data-density="compact"] { --d-py: 0.35rem; --d-px: 0.625rem;
--d-gap: 0.5rem; --d-card: 1rem; font-size: 0.875rem; } [data-
density="spacious"] { --d-py: 1.125rem; --d-px: 1.375rem; --d-gap: 1.5rem;
--d-card: 2rem; } /* ?? Audience Gating
???????????????????????????????????????????????????? */ /* Sections marked .aud-
manager are hidden for the executive audience. Sections marked .aud-engineer are
hidden for executive and manager. */ [data-audience="executive"] .aud-manager,
[data-audience="executive"] .aud-engineer { display: none; } [data-
audience="manager"] .aud-engineer { display: none; } /* ?? Base Styles
???????????????????????????????????????????????????? */ *, *::before, *::after
{ box-
sizing: border-box; } body { background: var(--bg-page); color: var(--text-1);
font-family: ui-sans-serif, system-ui, -apple-system, sans-serif; margin: 0;
padding: 0; transition: background 0.2s ease, color 0.2s ease; } main { max-
width: 64rem; margin: 0 auto; padding: 2.5rem 1.5rem; } /* ?? Controls Bar
???????????????????????????????????????????????????? */ .ctrl-bar { position:
sticky; top: 0; z-index: 20; background: var(--bg-card); border-bottom: 1px
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solid var(--border); padding: 0.5rem 1.5rem; display: flex; align-items: center; gap: 0.75rem; flex-wrap: wrap; box-shadow: var(--shadow); } .ctrl-brand { font-size: 0.7rem; font-weight: 700; letter-spacing: 0.15em; color: var(--accent); margin-right: auto; text-transform: uppercase; } .ctrl-group { display: flex; align-items: center; gap: 0.35rem; } .ctrl-label { font-size: 0.68rem; text-transform: uppercase; letter-spacing: 0.08em; color: var(--text-3); } .ctrl-select, .ctrl-btn { font-size: 0.78rem; padding: 0.2rem 0.5rem; border: 1px solid var(--border); border-radius: 0.375rem; background: var(--bg-muted); color: var(--text-1); cursor: pointer; transition: background 0.15s, border-color 0.15s; } .ctrl-select:hover, .ctrl-btn:hover { background: var(--accent-bg); border-color: var(--accent); color: var(--accent); } .ctrl-btn--active { background: var(--accent) !important; color: #fff !important; border-color: var(--accent) !important; } /* ?? Cards ?? */ .card { background: var(--bg-card); border: 1px solid var(--border); border-radius: 1rem; padding: var(--d-card); margin-bottom: 2rem; box-shadow: var(--shadow); } .card-title { font-size: 1.125rem; font-weight: 600; color: var(--text-1); margin: 0 0 1rem; } .card-subtitle { font-size: 0.95rem; font-weight: 600; color: var(--text-1); margin: 1.25rem 0 0.625rem; } /* ?? KPI Grid ?? */ .kpi-grid { display: grid; gap: var(--d-gap); grid-template-colum
ns: repeat(auto-fill, minmax(9rem, 1fr)); margin-top: 1.25rem; } .kpi-cell { background: var(--bg-muted); border-radius: 0.75rem; padding: var(--d-py) var(--d-px); } .kpi-label { font-size: 0.68rem; text-transform: uppercase; letter-spacing: 0.1em; color: var(--text-3); margin: 0; } .kpi-value { font-size: 1.1rem; font-weight: 600; color: var(--text-1); margin: 0.2rem 0 0; } /* ?? Tables ?? */ .tbl-wrap { border-radius: 0.75rem; overflow: hidden; border: 1px solid var(--border); margin-top: 1rem; } table { width: 100%; border-collapse: collapse; background: var(--bg-card); } thead tr { background: var(--bg-header); } thead th { padding: var(--d-py) var(--d-px); text-align: left; font-size: 0.7rem; font-weight: 600; text-transform: uppercase; letter-spacing: 0.06em; color: var(--text-2); } tbody td { padding: var(--d-py) var(--d-px); color: var(--text-2); border-top: 1px solid var(--border); font-size: 0.875rem; } /*

Row-level status tints driven by tr.pass / tr.warn / tr.fail classes. These interact with the CSS token system so they invert correctly in dark mode without any extra JavaScript. */ tbody tr.pass td { background: var(--pass-bg); color: var(--pass-text); } tbody tr.warn td { background: var(--warn-bg); color: var(--warn-text); } tbody tr.fail td { background: var(--fail-bg); color: var(--fail-text); } /* ?? Status Labels ?? */ .status-pass { color: var(--pass-text); font-weight: 700; font-size: 0.7rem; text-transform: uppercase; letter-spacing: 0.08em; } .status-warn { color: var(--warn-text); font-weight: 700; font-size: 0.7rem; text-transform: uppercase; letter-spacing: 0.08em; } .status-fail { color: var(--fail-text); font-weight: 700; font-size: 0.7rem; text-transform: uppercase; letter-spacing: 0.08em; } /* ?? Badge ?? */ .badge { display: inline-block; border-radius: 999px; padding: 0.15 rem 0.625rem; font-size: 0.68rem; font-weight: 600; letter-spacing: 0.06em; } .badge-accent { background: var(--accent-bg); color: var(--accent); } /* ?? Inline code ?? */ code { background: var(--bg-muted); border-radius: 0.25rem; padding: 0.1em 0.35em; font-size: 0.85em; color: var(--accent); } BLSN Theme ? Light ? Dark ? Auto Density Compact Comfortable Spacious Audience Executive Manager Engineer ? Wave 15 ? No Drift or Regression Detected No drift detected (Z=0.00, threshold=±3.0?). BLSN Synthetic Validation Report Theme light · Density comfortable · Audience engineer Vehicle P3 Orion Cooling Mode helium_closed_loop Nominal Flow 110 g/s Limits 80?140 g/s Sanity Check Telemetry Status Message PASS [PASS] PASS | Nominal mass flow is bounded by configured limits System Verification: Claimed vs. Actual Claimed Limits: 0 | Actually Tested: 0 ? STATUS: UNKNOWN Total Runtime 0.00s Pass Ratio 0.0% Fail Ratio 0.0% Skip Ratio 0.0% Tuple Execution Telemetry Targets (seconds) Metric Target Warn Fail ci_total_runtime_s 120.0 180.0 240.0 pytest_runtime_s 15.0 25.0 35.0 pipeline_runtime_s 5.0 10.0 15.0 yaml_json_lint_runtime_s 3.0 6.0 9.0 Analytics Dashboard Executive View Engineering View System Priority & Bottleneck Ranking (Bradley-Terry) Pairwise probability model: P(i > j) = p_i / (p_i + p_j) Global Repo Ranking Rank Artifact BT Strength Win Prob vs Top 1 docs/index.html 2.533414 50.00% 2

docs/_includes/metric_card.html 2.317756 47.78% 3 docs/presentation.html 2.309540 47.69% 4 docs/_includes/plotly_render.html 2.246689 47.00% 5 docs/_layouts/default.html 2.161844 46.04% 6 docs/blsn_report.pdf 0.795024 23.89% 7 docs/visuals/plotly_payloads.json 0.481794 15.98% 8 docs/experimental_design_matrix.json 0.417307 14.14% 9 docs/historical_telemetry_ledger.json 0.409257 13.91% 10 docs/federation/federation_cherry_pick.json 0.382788 13.13% Inter-Category Ranking R
ank Category Win Prob vs Top 1 html 50.00% 2 json 15.19% 3 md 10.52% 4 pdf 6.43% 5 yaml 3.18% 6 yml 2.87% QPLANT Requirements Ranking Bridge Rank Requirement Win Prob vs Top 1 ci_total_runtime_s 50.00% 2 pytest_runtime_s 50.00% 3 pipeline_runtime_s 50.00% 4 yaml_json_lint_runtime_s 50.00% Wave 12: SPC & Telemetry Stats Runtime Drift 0.0000s Prev Moving Avg 0.0000s Total Runs 2 Success Rate 100.0% DoE Factors ? A: python_version · B: os_type · C: load_parameter Principal Component Analysis (PCA Proof) Principal Component Eigenvalue Explained Variance Highest Weight Variable PC1 0.000000 0.00% total_tests PC2 0.000000 0.00% skip_rate Metric Markdown Slugs Ci Total Runtime S Pytest Runtime S Pipeline Runtime S Yaml Json Lint Runtime S Ci Total Runtime S Metric ID: `ci_total_runtime_s` Target Threshold: **120.0s** Warn Threshold: **180.0s** Fail Threshold: **240.0s** Pytest Runtime S Metric ID: `pytest_runtime_s` Target Threshold: **15.0s** Warn Threshold: **25.0s** Fail Threshold: **35.0s** Pipeline Runtime S Metric ID: `pipeline_runtime_s` Target Threshold: **5.0s** Warn Threshold: **10.0s** Fail Threshold: **15.0s** Yaml Json Lint Runtime S Metric ID: `yaml_json_lint_runtime_s` Target Threshold: **3.0s** Warn Threshold: **6.0s** Fail Threshold: **9.0s** Mermaid System Flow Diagram graph LR; SSOT[SSOT Config] --> PIPELINE[Pipeline Execution]; PIPELINE --> MARKDOWN[Markdown Slug Pages]; PIPELINE --> INDEX[Index HTML]; PIPELINE --> SLIDES[Slides HTML]; INDEX --> PDF[PDF Export]; /* ?? BLSN Report UI Controls ?????????????????????????????????????? All user preferences are persisted to localStorage under the "blsn_" namespace so they survive hard reloads without server round-trips. Theme: light | dark | auto "auto" resolves to light or dark via prefers-color-scheme and keeps the attribute updated when the OS preference changes (e.g. switching between light and dark system themes). Density: compact | comfortable | spacious Controls the --d-py / --d-px / --d-gap / --

d-card CSS tokens that scale table cell padding and card gutters without reflowing the page. Audience: executive | manager | engineer CSS [data-audience] selectors hide .aud-manager and .aud-engineer sections for less technical readers. Changing this value simply updates the data attribute; no DOM nodes are added or removed. Plotly charts are re-rendered via Plotly.react() rather than Plotly.newPlot() when the theme changes so the existing DOM element is reused and transitions feel instant. ?? */ (function () { 'use strict'; const PREF_NS = 'blsn_'; const getPref = (k, def) => localStorage.getItem(PREF_NS + k) || def; const setPref = (k, v) => localStorage.setItem(PREF_NS + k, v); const root = document.documentElement; /* Resolve the "auto" theme sentinel using the OS preference. */ function resolveTheme(raw) { if (raw !== 'auto') return raw; return window.matchMedia('(prefers-color-scheme: dark)').matches ? 'dark' : 'light'; } /* Apply a data-attribute and keep its matching in sync. */ function applyAttr(attr, selectId, value) { root.setAttribute('data-' + attr, value); const el = document.getElementById(selectId); if (el) el.value = value; setPref(attr, value); } /* ?? Plotly theme-aware rendering ?????????????????????????????? */ let _payload = null; /* cached payload to allow re-render on theme change */ /* Patch a Plotly chart layout for dynamic font/grid colours. Backgrounds stay transparent (set by Python); only text and grid colours are adjusted so light/dark switching works seamlessly. */ function patchLayout(chart, themeRaw) { const isDark = resolveTheme(themeRaw) === 'dark'; const gridColor = isDark ? '#334155' : '#e2e8f0'; const fontColor = isDark ? '#94a3b8' : '#475569'; return Object.assign({}, chart.layout || {}, { paper_bgcolor: 'rgba(0,0,0,0)', plot_bgcolor: 'rgba(0,0,0,0)', font: { color: fontColor }, xaxis: Object.assign({}, (chart.layout || {}).xaxis, { gridcolor: gridColor, zerolinecolor: gridColor }), yaxis: Object.assign({}, (chart.layout
||
{}).yaxis, { gridcolor: gridColor, zerolinecolor: gridColor }), }); } /* Re-render both Plotly charts with the resolved theme. Reads from audience-structured payload (executive / engineering). */ function rerenderPlotly(themeRaw) { if (!window.Plotly || !_payload) return; const audience = _payload.audience || {}; const tel = (audience.executive || {}).telemetry_combo || {}; const pca = (audience.engineering || {}).pca_bottlenecks || {}; Plotly.react('plotly-telemetry-chart', tel.data ||

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[], patchLayout(tel, themeRaw)); Plotly.react('plotly-pca-chart', pca.data ||
[], patchLayout(pca, themeRaw)); } /* ?? Chart-view audience tab toggle
?????????????????????????????? */ window.switchChartView = function (view) {
const viewExec = document.getElementById('view-executive'); const viewEng =
document.getElementById('view-engineering'); const btnExec =
document.getElementById('btn-exec'); const btnEng =
document.getElementById('btn-eng'); if (!viewExec || !viewEng) return; const
isExec = view === 'executive'; viewExec.style.display = isExec ? 'block' :
'none'; viewEng.style.display = isExec ? 'none' : 'block'; if (btnExec)
btnExec.classList.toggle('ctrl-btn--active', isExec); if (btnEng)
btnEng.classList.toggle('ctrl-btn--active', !isExec); /* Trigger resize so
Plotly fills a div that was previously hidden. */ window.dispatchEvent(new
Event('resize')); }; /* ?? Public setters wired to onchange handlers
????????????????? */ window.setTheme = function (v) { /* Apply the resolved
value to data-theme but store the raw choice (including "auto") so the select
always reflects user intent. */ root.setAttribute('data-theme',
resolveTheme(v)); const el = document.getElementById('ctrl-theme'); if (el)
el.value = v; setPref('theme', v); rerenderPlotly(v); }; window.setDensity = (v)
=> applyAttr('density', 'ctrl-density', v); window.setAudience = (v) =>
applyAttr('audience', 'ctrl-audience', v); /* ?? Initialise from saved
preferences (fall back to SSOT defaults) */ const initTheme =
 getPref('theme', 'light'); const initDensity = getPref('density',
'comfortable'); const initAudience = getPref('audience', 'engineer'); /* Apply
theme without going through the setter so the select shows "auto" rather than
the resolved value when stored as "auto". */ root.setAttribute('data-theme',
resolveTheme(initTheme)); const themeEl = document.getElementById('ctrl-theme');
if (themeEl) themeEl.value = initTheme; applyAttr('density', 'ctrl-density',
initDensity); applyAttr('audience', 'ctrl-audience', initAudience); /* ?? Plotly
payload fetch ?????????????????????????????????????? */ const emptyLayout = ()
=> ({ paper_bgcolor: 'rgba(0,0,0,0)', plot_bgcolor: 'rgba(0,0,0,0)', xaxis: {
visible: false }, yaxis: { visible: false }, annotations: [{ text: 'No Plotly
payload available', showarrow: false, xref: 'paper', yref: 'paper', x: 0.5, y:
0.5, }], }); fetch('visuals/plotly_payloads.json') .then((r) => (r.ok ? r.json()
: Promise.reject(new Error('HTTP ' + r.status)))) .then((payload) => { _payload
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= payload; rerenderPlotly(getPref('theme', 'light')); }) .catch(() => { const el
= emptyLayout(); Plotly.newPlot('plotly-telemetry-chart', [], el);
Plotly.newPlot('plotly-pca-chart', [], el); }); /* Keep "auto" theme in sync
when the OS preference changes at runtime (e.g. user switches system dark mode
while the tab is open). */ window.matchMedia('(prefers-color-scheme:
dark)').addEventListener('change', () => { const stored = getPref('theme',
'light'); if (stored === 'auto') { root.setAttribute('data-theme',
resolveTheme('auto')); rerenderPlotly('auto'); } }); })();